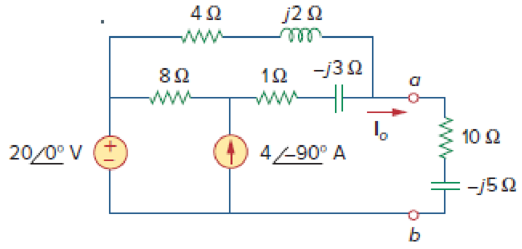
Chapter 10, Problem 35P

Problem

Calculate **I**o in Fig. 10.30 (for Practice Prob. 10.10) using mesh analysis.

Figure 10.30:



Step-by-step solution

Step 1 of 6

Refer to Figure 10.30 in the textbook.

Draw the modified circuit with the loop current notations.

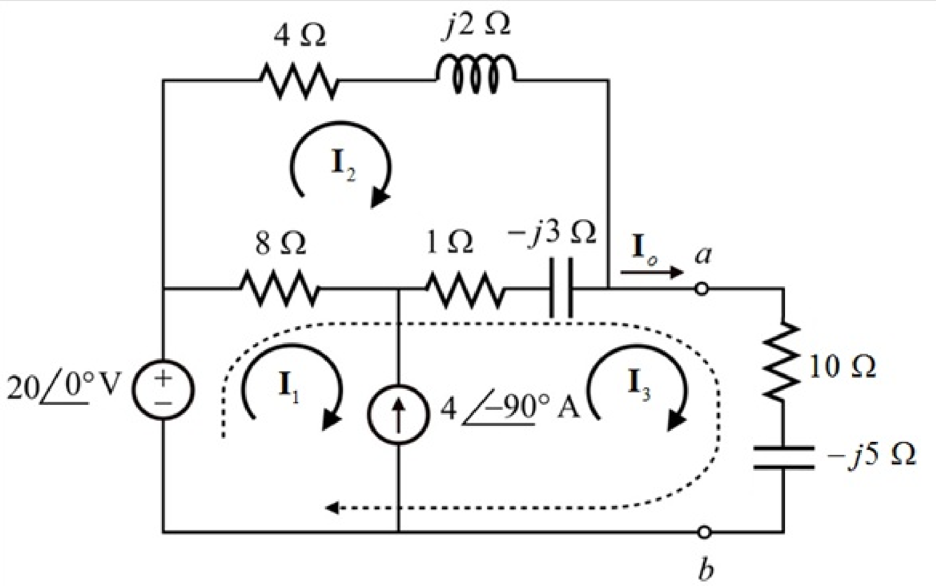


Figure 1

Step 2 of 6

Apply Kirchhoff's voltage law for the mesh-2.

$$8(\mathbf{I}_2-\mathbf{I}_1)+4\mathbf{I}_2+j2\mathbf{I}_2-j3(\mathbf{I}_2-\mathbf{I}_3)+1(\mathbf{I}_2-\mathbf{I}_3)=0$$
$$8(\mathbf{I}_2-\mathbf{I}_1)+(4+j2)\mathbf{I}_2+(1-j3)(\mathbf{I}_2-\mathbf{I}_3)=0$$
$$-8\mathbf{I}_1+(8+4+j2+1-j3)\mathbf{I}_2-(1-j3)\mathbf{I}_3=0$$
$$-8\mathbf{I}_1+(13-j)\mathbf{I}_2-(1-j3)\mathbf{I}_3=0 \quad \ldots\ldots. (1)$$

Apply Kirchhoff's voltage law for super mesh.

$$-20\angle 0°+8(\mathbf{I}_1-\mathbf{I}_2)+1(\mathbf{I}_3-\mathbf{I}_2)-j3(\mathbf{I}_3-\mathbf{I}_2)+10\mathbf{I}_3-j5\mathbf{I}_3=0$$
$$-20+8(\mathbf{I}_1-\mathbf{I}_2)+(1-j3)(\mathbf{I}_3-\mathbf{I}_2)+(10-j5)\mathbf{I}_3=0$$
$$8\mathbf{I}_1-(8+1-j3)\mathbf{I}_2+(1-j3+10-j5)\mathbf{I}_3=20$$
$$8\mathbf{I}_1-(9-j3)\mathbf{I}_2+(11-j8)\mathbf{I}_3=20 \quad \ldots\ldots. (2)$$

Step 3 of 6

From the mesh-1 and mesh-3,

$$\mathbf{I}_3-\mathbf{I}_1=4\angle -90°$$
$$-\mathbf{I}_1+\mathbf{I}_3=-j4 \quad \ldots\ldots. (3)$$

Step 4 of 6

Solve equations (1), (2) and (3) using Crammer's rule.

$$\begin{bmatrix} -8 & (13-j) & -(1-j3) \\ 8 & -(9-j3) & (11-j8) \\ -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \\ \mathbf{I}_3 \end{bmatrix}=\begin{bmatrix} 0 \\ 20 \\ -j4 \end{bmatrix}$$

Determine the value of Δ.

$$\Delta=\begin{vmatrix} -8 & (13-j) & -(1-j3) \\ 8 & -(9-j3) & (11-j8) \\ -1 & 0 & 1 \end{vmatrix}$$
$$=-8\left[-(9-j3)-0\right]-(13-j)\left[8+(11-j8)\right]-(1-j3)\left[0-(9-j3)\right]$$
$$=8(9-j3)+(-13+j)(19-j8)+(-1+j3)(-9+j3)$$
$$=(72-j24)+(-239+j123)+(-j30)$$
$$\Delta=-167+j69$$

Step 5 of 6

Determine the value of Δ_3.

$$\Delta_3=\begin{vmatrix} -8 & (13-j) & 0 \\ 8 & -(9-j3) & 20 \\ -1 & 0 & -j4 \end{vmatrix}$$
$$=-8\left[(j4)(9-j3)-0\right]-(13-j)\left[8(-j4)+20\right]+0$$
$$=-8(12+j36)+(-13+j)(20-j32)$$
$$=(-96-j288)+(-228+j436)$$
$$\Delta_3=-324+j148$$

Step 6 of 6

Determine the value of current, $\mathbf{I}_o$.

$$\mathbf{I}_o = \mathbf{I}_3$$

$$\mathbf{I}_o = \frac{\Delta_3}{\Delta}$$

Substitute $-324 + j148$ for Δ_3 and $-167 + j69$ for Δ in the equation.

$$\begin{aligned}\mathbf{I}_o &= \frac{-324 + j148}{-167 + j69} \\ &= 1.97 - j0.0723 \\ &= 1.9713\angle -2.10^\circ \text{ A}\end{aligned}$$

Thus, the value of mesh current, $\mathbf{I}_o$ is $\boxed{1.9713\angle -2.10^\circ \text{ A}}$.